

Predicting Food Delivery Time

Complete workflow — objective, data, cleaning & modelling

1. Objective

Build a regression model that predicts how long a delivery will take (in minutes) using four measurable inputs: distance travelled, number of items, traffic level, and food preparation time.

2. Dataset overview

1,020

delivery records

4

input features

~94 min

avg. delivery time

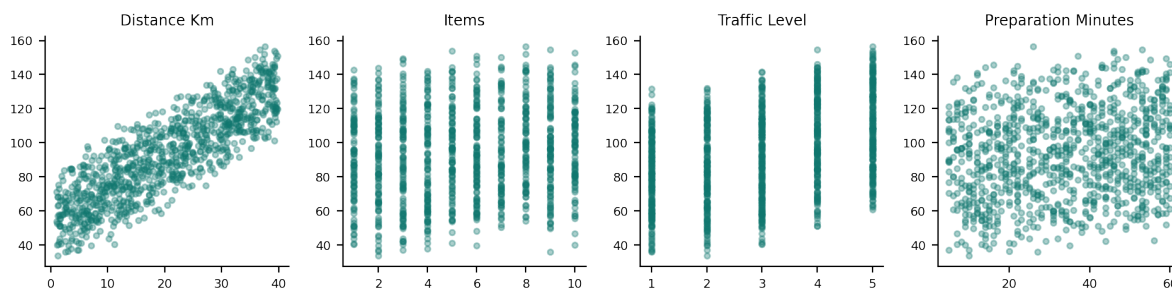
80 / 20

train/test split

3. Data cleaning

- Negative distance_km values were found and treated as invalid (impossible measurement).
- Negative items values were treated as invalid (an order cannot contain negative items).
- traffic_level should range 1–5, but a few rows showed 10 — treated as invalid.
- All remaining missing values were filled using the median of each column (robust to outliers).

4. Feature relationships with delivery time



5. End-to-end workflow



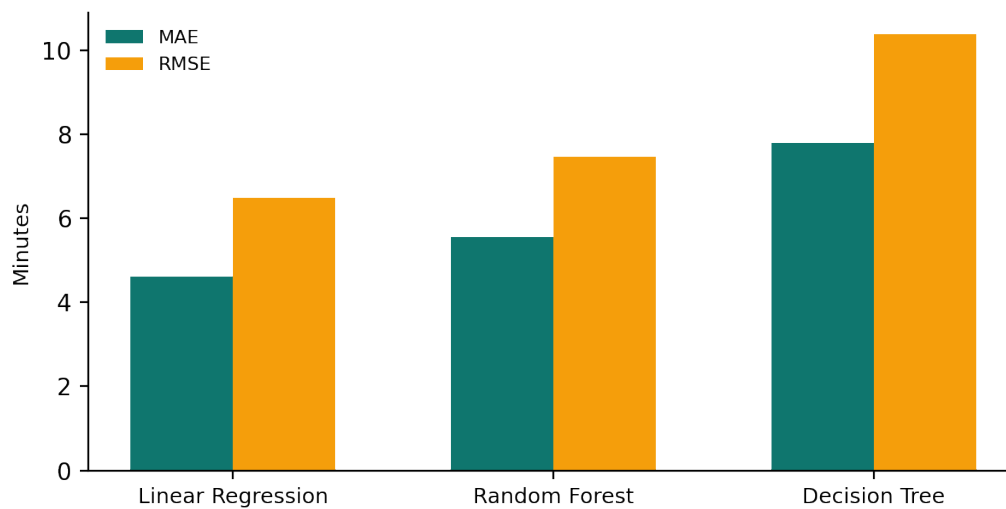
6. Tools used

Python, pandas, NumPy, scikit-learn (pipelines, imputation, scaling, regressors), matplotlib and seaborn for visualization.

Model Results & Outcome

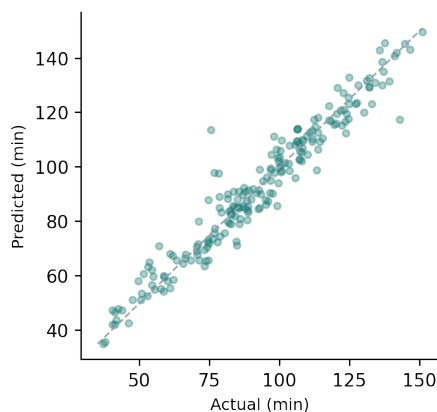
Comparison, evaluation and final takeaways

7. Model comparison

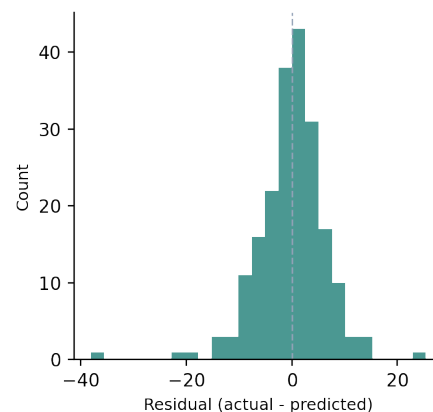


Lower MAE / RMSE = better. Linear Regression has the lowest error of the three models.

8. Best model diagnostics — Linear Regression



Predicted vs actual delivery time



Residual distribution

9. Key results

- Linear Regression was the best-performing model: MAE of 4.6 minutes, RMSE of 6.5 minutes, and an R^2 of 0.94 — meaning it explains about 94% of the variation in delivery time.
- Random Forest was close behind ($R^2 = 0.92$), while the single Decision Tree performed noticeably worse ($R^2 = 0.84$), suggesting the relationship between features and delivery time is largely linear.
- Data cleaning mattered: invalid values (negative distance/items, out-of-range traffic level) and missing entries were corrected before training to avoid distorted predictions.

10. Outcome

The trained pipeline predicts delivery time within about 4-5 minutes on average from just four order details, and can be reused directly on new orders to power live ETA estimates.